Tesla Shelf | spec sheet

Tesla Lab Product Design Engine - numbers computed in Python, not guessed

A bookshelf that powers everything sitting on or near it, no cables, no pads.

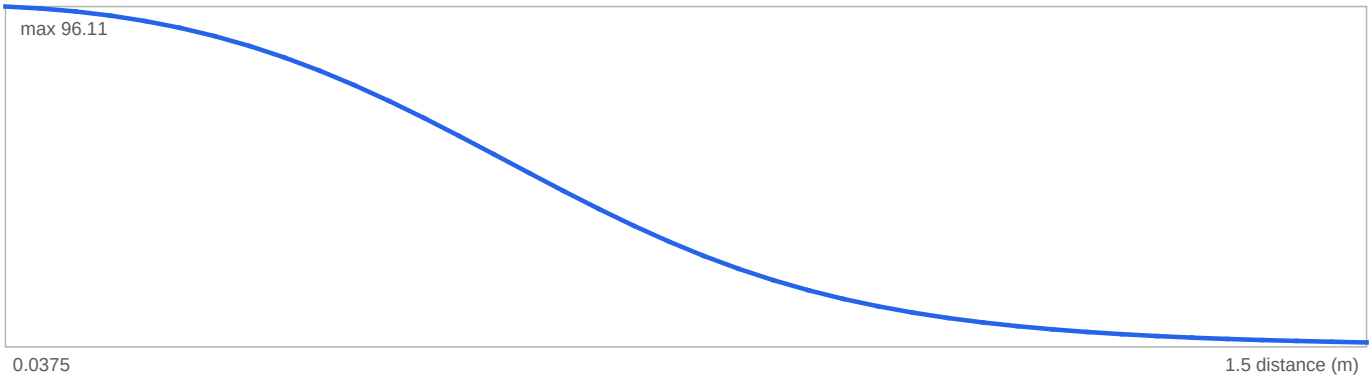
Derived from the claim that WORKS: Electric power can be sent through the air between two tuned (resonant) coils.



Key numbers

- 10 W into the load at 0.5 m with 63% coil-to-coil efficiency
- 50 cm 2-turn copper coil at 6.78 MHz, Q ~ 717
- Efficiency halves by 0.75 m and is below 10% beyond 1.01 m
- Exposure screen: 10 W fails the whole-body level on-axis at 0.5 m; passes beyond 0.85 m (or at <=0.6 W)
- Prototype parts ~ $380 (estimate)

coil-to-coil efficiency (%) vs distance (m)



Requirements

Power into the receiver load	**10 W**	req
Design distance (face to face)	**0.5 m**	req
Operating frequency	**6.78 MHz**	req

6.78 MHz ISM band used by AirFuel Resonant

Transmitter coil

Diameter	**0.5 m**	req
Turns	**2**	calc
best efficiency at the design distance, wire < lambda/10		
Conductor	**6 mm copper tube**	req
Wire length	**3.14 m**	calc
lambda = 44.2 m, so the lumped model holds		
Inductance	**4.79 uH**	calc
Tuning capacitor	**115 pF**	calc
727 V rms across it at full power; spec >= 2500 V, low-loss (C0G/porcelain)		
Quality factor Q	**717**	calc
copper 0.147 ohm + radiation 0.0013 ohm + capacitor ESR 0.136 ohm (Q_cap assumed)		

Receiver coil

Diameter	**0.15 m**	req
Turns	**6**	calc
Inductance	**9.07 uH**	calc
Tuning capacitor	**60.8 pF**	calc
Quality factor Q	**590**	calc

Link at design distance

Coupling k	**0.0066**	calc
Figure of merit k*sqrt(Q1*Q2)	**4.28**	calc
Coil-to-coil efficiency (optimal load)	**62.9 %**	calc
Optimal load resistance	**2.88 ohm**	calc
Amplifier output power	**15.9 W**	calc
Wall power (amplifier eta assumed)	**18.7 W**	calc
amplifier efficiency 85% assumed (GaN class-E/D)		
Transmitter coil current	**3.56 A rms**	calc

Range

Efficiency halves at	**0.75 m**	calc
Efficiency below 10% beyond	**1.01 m**	calc

Exposure screen

Max power passing the whole-body screen at design distance	**0.65 W**	calc
H ~ sqrtP; on-axis peak field, conservative		
Max power passing the local-exposure screen	**3.22 W**	calc
Keep-out distance on the coil axis at full power	**0.85 m**	calc
whole-body reference level		

Exposure / safety screen (approximate)

FAIL	H-field at the receiver (0.5 m, on-axis): 1.275 vs limit 0.324 A/m. ICNIRP 2020, Table 5 (whole body, 30-min avg): H = 2.2/f_MHz A/m, 0.1-30 MHz
FAIL	H-field at the receiver (0.5 m, on-axis): 1.275 vs limit 0.723 A/m. ICNIRP 2020, Table 6 (local, 6-min avg): H = 4.9/f_MHz A/m, 0.1-30 MHz
PASS	H-field at the receiver (0.5 m, on-axis): 1.275 vs limit 21 A/m. ICNIRP 2010, Table 4 (nerve stimulation): H = 21 A/m, 3 kHz-10 MHz

Field falls below this limit 0.85 m from the coil face (on-axis peak; body-averaged field is lower). Screening estimate, not a compliance test.

Bill of materials (estimates)

6.78 MHz GaN amplifier board (class-E/D, ~20-30 W)	**1 x $120.00**	$120
Copper tube for tx coil (m)	**3.6 x $6.00**	$22

Tx tuning capacitors (>=2500 V, low-loss)	**2 x $18.00**	$36
Receiver coil + capacitor (per device)	**3 x $6.00**	$18
Receiver rectifier + DC-DC (per device)	**3 x $9.00**	$27
Controller with BLE (power negotiation, foreign-object cut-off)	**1 x $12.00**	$12
24 V power supply, 60 W	**1 x $25.00**	$25
Shelf body, back panel, ferrite-free mounting	**1 x $80.00**	$80
Near-field probe / test equipment share	**1 x $40.00**	$40

Total (estimate): $380

Rough estimates from typical 2025-26 distributor / retail list prices for one-off prototype quantities, not quotes. Verify before you buy.

Honest limits

About 10 W per device at half a metre, and efficiency halves by about 0.75 m (see the spec sheet). At full power the exposure screen says keep people roughly 0.85 m from the coil's axis, or run below about 0.6 W. It will never power a house across town.

Caveats

- Coaxial, aligned coils; lateral offset or tilt lowers k and efficiency.
- Quasi-static model (coils much smaller than the 44 m wavelength); no ferrite or nearby metal.
- Capacitor Q, proximity factor and amplifier efficiency are assumptions; measure them on the prototype.
- The exposure screen compares the on-axis peak H-field with ICNIRP reference levels. Real compliance needs measurement or SAR/induced-field simulation against the basic restrictions.

Render prompt (generated from these numbers)

Photoreal product render, 3/4 view, warm studio light, clean background, no text. A minimalist oak bookshelf about 1.2 m wide; a single 50 cm diameter circular coil of 2 turns of 6 mm brushed-copper tube is inlaid flush in the back panel, centred behind the middle shelf. Two small cylindrical tuning capacitors and a palm-sized amplifier box sit at the coil's lower edge. On the shelf, within 50 cm of the coil face, a desk lamp whose round base hides a 15 cm receiver coil glows softly; a phone and a small plant sensor sit nearby. Faint pale-blue magnetic field lines loop through the coil and fade out within about 0.75 m. A subtle floor marking shows a 0.85 m keep-clear arc in front of the coil. Shallow depth of field.

References

- Kurs et al., Wireless power transfer via strongly coupled magnetic resonances, Science 317 (2007) https://pubmed.ncbi.nlm.nih.gov/17556549
- ICNIRP (2020) RF guidelines, Tables 5-6 https://www.icnirp.org/cms/upload/publications/ICNIRPrfgdl2020.pdf
- ICNIRP (2010) low-frequency guidelines, Table 4 https://www.icnirp.org/cms/upload/publications/ICNIRPLFgdl.pdf
- Tesla, US 645,576 / 649,621 (1900) - the original tuned-circuit transmission patents https://patents.google.com/patent/US645576A/en